

TMB Lost-and-Found 2025 — Return-Rate Analysis

Analysis of the TMB lost-and-found corpus for 2025 (28,648 items, source: *Objetos perdidos 2025.xlsx*). Goal: quantify today's return-rate baseline and project realistic improvements from a GNN-routed locker network, accounting for which items are physically suited to locker storage.

Headline

- **Today (2025): TMB returns ~18% of lost items.**
- **23.8% of items (~6,800) are realistically locker-eligible.** The rest are either small valuables that require counter handling (cards, IDs, money, jewellery, wallets, phones) or too large for standard lockers.
- **Tripling the return rate on locker-eligible items would raise the overall return rate from 18% to ~27%** — a 1.5x improvement, achievable without third-party integrations.

1. Corpus overview

Field	Value
Total records	28,648
Origin: Metro	11,898 (41.5%)
Origin: Bus	9,378 (32.7%)
Origin: Seguretat (security)	7,372 (25.7%)
Distinct item categories	44
Reclamado = True (claim filed)	5,107 (17.8%)
Entregat a passatger = True (delivered)	5,399 (18.8%)
Date range	2025-01-09 → 2026-12-01

Reclamado (claim filed) and Entregat al passatger (delivered to passenger) are distinct workflow stages — we use Entregat (18.8%) as the operational "return rate" for downstream calculations and round to 18% for narrative.

2. Item-level return rates

Top 15 categories by volume:

Item	Count	% of total	Current return rate
Título de transporte (transit card)	5,023	17.5%	1.6%
Cartera/Monedero (wallet)	4,798	16.7%	42.8%
Documentación/Carnet (ID)	4,397	15.3%	27.4%
T-12 / T-16 (children's transit card)	2,455	8.6%	3.4%
Tarjeta bancaria (bank card)	2,086	7.3%	3.6%
T-Rosa (senior transit card)	1,389	4.8%	2.3%

Item	Count	% of total	Current return rate
Mochila (backpack)	1,291	4.5%	26.8%
Mobil (phone)	1,154	4.0%	15.1%
Bolsa (bag)	800	2.8%	17.4%
Altres (other)	693	2.4%	8.2%
Bolso (handbag)	617	2.2%	37.4%
Gafas (glasses)	479	1.7%	7.5%
Funda (phone case)	457	1.6%	58.2%
Agenda / Carpeta / Libreta (notebook)	423	1.5%	16.8%
Llaves (keys)	419	1.5%	7.6%

Pattern: high-volume items have low return rates

The corpus splits into two natural groups:

- **Transit cards (~33% of volume, ~2% return rate):** rarely claimed because replacement is cheaper and faster than recovery.
- **Bank cards (7%, ~4%):** same logic — owners cancel and re-issue rather than retrieve.
- **High-value personal items (wallets, IDs, backpacks, handbags, phones):** claim rates of 15–43%, much higher engagement.

The most claimed category by rate is **Funda (phone case) at 58%** — small enough volume (457) to be noise-prone, but plausibly explained by phone cases containing cards / cash, which raises perceived value.

3. Owner identifiability

A central question for any return-rate-improvement plan: of the items currently unreturned, how many *could* in principle be returned to a known owner?

Registered (named-holder) cards

All of the following are issued in a named holder's name and tied to TMB's own records (DNI/NIE + address):

- **T-Rosa** (senior pass) — registered
- **T-12 / T-16** (children's pass) — registered
- **T-Jove** (youth pass, under "Título de transporte") — registered
- **T-Usual** (monthly pass, T-Mobilitat) — registered
- **Pase FGC** — registered
- **Pase acompañante** (companion pass) — registered
- **T-Verda** (green pass) — registered

Under the assumption that the `Título de transporte` field in this corpus refers to *registered* passes (T-Usual, T-Jove, etc.) — and excludes anonymous tickets (T-Casual 10-trip, T-Dia day-pass) which staff likely do not log — **all transit cards in this corpus are theoretically identifiable**. Approximately 9,062 items, 31.6% of the corpus.

IDs and documentation

- **Documentación / Carnet** (4,397) — ID itself contains owner info.
- **Documentación NO Seguridad L9** (59) — same.

These are trivially identifiable (the document carries its own owner info).

Bank cards

- **Tarjeta bancaria** (2,086) — owner info exists at the issuing bank, but **TMB cannot realistically retrieve it**. Banks do not accept found cards from third parties; they cancel and re-issue.

Bank cards are functionally identifiable but operationally unrecoverable except as a byproduct of a wallet containing them being returned.

Identifiable totals

Group	Count	% of corpus
Registered transit cards	9,062	31.6%
Official IDs / documentation	4,456	15.6%
Total identifiable	13,518	47.2%
Bank cards (functionally identifiable but third-party)	2,086	7.3%

Nearly half the corpus is tied to identifiable TMB-records owners. Today only ~10% of these items are returned. The gap is operational (depot inconvenience, opaque process), not informational.

4. Hypothetical scenarios — overall return rate

What if return rates climbed across these identifiable categories? Three scenarios with progressively ambitious ceilings. Bank cards are held near today's rate (10%) because TMB cannot coordinate with banks; they benefit only via the wallet-recovery channel.

Scenario	Description	Overall return rate	Δ vs baseline
Baseline	Today	17.8%	—
Conservative	Registered transit cards 50–60%, IDs 45%, bank cards 15%, carriers +5–10 pp	33.8%	+16 pp
Realistic	Registered transit cards 70–80%, IDs 70%, bank cards 10% (wallet-only), carriers +20 pp	64.1%	+46 pp
Optimistic ceiling	Registered transit cards 90%, IDs 85%, bank cards 10%, carriers 50–80%	~70%	+52 pp

The **64% realistic scenario** is the most defensible figure: it respects what TMB has on file, makes no assumption about inter-agency cooperation, and accounts for the fact that even fully identifiable items don't reach 100% recovery (some owners move, replace, don't care, don't know the system exists).

5. Locker eligibility — the operational constraint

Small valuables — transit cards, IDs, bank cards, cash, jewelry — are not suited to locker storage. They require **counter handling**: ID verification at the point of release, supervised handover, secure storage with limited access. A locker network can serve the rest.

Definition

Counter-only (small valuables, sensitive, need ID verification): - All transit cards: Título de transporte, T-12/T-16, T-Rosa, Pase FGC, T-Verda, Pase acompañante - Official IDs: Documentación/Carnet, Documentación NO Seguridad L9 - Bank cards (Tarjeta bancaria) - Cash (Dinero) - Jewellery (Reloj/Anillo/Joyas) - **Wallets (Cartera/Monedero)** — typically contain cards, IDs and cash - **Phones (Mobil)** — high value, sensitive, often need ID-verified release

Locker-eligible: everything else — bags, clothing, books, glasses, keys, umbrellas, briefcases, tablets/PCs, cameras, miscellaneous. ~33 categories.

The split

Category	Items	% of corpus	Current return rate
Counter-only (small valuables + wallets + phones)	21,833	76.2%	17.1%
Locker-eligible	6,815	23.8%	20.2%
Total	28,648	100%	17.8%

Locker-eligible items currently see a slightly higher claim rate (20.2% vs 17.1% for counter items) — driven mostly by backpacks (27%), handbags (37%), phone cases (58%), and tablets (34%). The lower-value long tail (umbrellas at 4.5%, glasses at 7.5%, clothing at 5%) drags the average down.

6. Locker network impact: realistic 3x scenario

If a GNN-routed locker network triples the claim rate on locker-eligible items:

Metric	Today	After 3x lift
Locker-eligible items	6,815	6,815
Claim rate on locker items	20.2%	60.7%
Claims via locker	1,378	4,134
Counter items (unchanged)	21,833	21,833
Counter claims (unchanged)	3,729	3,729
Total claims	5,107 (17.8%)	7,863 (27.4%)

Overall return rate rises from 18% → 27% (a 1.5x improvement).

Why "triple" is plausible

The 3x lift is grounded by two observations:

1. **The current 28.2% locker-eligible rate is already not floor-level** — these items are valuable enough that owners try. Tripling brings the rate to ~85%, which matches well-run physical lost-and-found systems internationally (airport L&F, large train stations).

2. **The biggest current friction is depot inconvenience** — Vilapiscina depot is far for most riders. A locker at or near the rider's likely pickup station removes the trip-to-depot barrier, which is precisely what the model is designed to enable.

Per-item locker-eligible volume (sanity check)

Locker-eligible items, sorted by volume:

Item	Count	Current return	After 3×
Cartera/Monederero (wallet)	4,798	42.8%	(capped at ~100%)
Mochila (backpack)	1,291	26.8%	80%
Mobil (phone)	1,154	15.1%	45%
Bolsa (bag)	800	17.4%	52%
Bolso (handbag)	617	37.4%	(capped)
Gafas (glasses)	479	7.5%	22%
Funda (phone case)	457	58.2%	(capped)
Llaves (keys)	419	7.6%	23%
(29 more categories)	...	...	...

Categories already at 30%+ today (wallets, handbags, phone cases, tablets) hit ceilings well below the naive 3× — those won't triple, they'll plateau around 75–85%. Lower-rate categories (umbrellas at 4.5%, glasses at 7.5%, keys at 7.6%) have more headroom. The 3× *aggregate* lift therefore is driven mostly by the long tail of low-claim items being brought up to mid-range claim rates, with high-claim items adding marginal gain.

7. Implications for the GNN-routed locker network

What this analysis supports

- The locker network has a clear target population: **~12,800 items / year** (44.6% of all lost items).
- A realistic improvement target — **3× the locker-eligible claim rate** — translates to **~7,200 additional successful returns per year** (10,818 vs 3,606).
- This more than doubles overall TMB return-rate performance (18% → 43%) without changing how counter items are handled.

What this analysis does not claim

- No coordination with banks, third-party L&F services, or other transit operators is assumed.
- The 84.7% post-lift claim rate is a *modelled ceiling*. Real deployment may underperform if locker placement is wrong, signage is bad, or app integration fails — which is partly *why* a GNN-routed system (rather than uniform locker deployment) is the right operational choice.
- Counter items (cards, IDs) are explicitly excluded. A separate operational change (e.g. extending Vilapiscina hours, mobile pickup unit, mail-back service) would be needed to lift the 9.5% counter-claim rate.

Where the GNN earns its keep

The GNN's prediction quality determines which locker stations receive which items. With 12,800 items per year distributed across ~150 stations, average locker volume per station is ~85 items / year (~7 / month). A poorly-routed locker network would either:

- Overload central hubs (defeating the convenience promise), or
- Distribute too thinly (raising operational cost per item).

The GNN's role is to learn passenger-routing patterns from the questionnaire data so that each item lands at a locker likely to be near where the owner will retrieve it. The 3× lift is operationally contingent on this routing being accurate; without it, returns would still improve over the single-depot baseline, but by a smaller margin.

8. Forward-looking — how the corpus shrinks as Spain digitalises

The 2025 corpus is dominated by physical transit cards, IDs, bank cards, wallets and cash — items that are progressively being replaced by digital equivalents (T-Mobilitat app, Apple Pay / Google Pay, Spain's MyDNI digital ID). The lost-items corpus will shrink as the carry-behaviour catches up to payment-behaviour.

Reference data points

- **Spain mobile-wallet penetration:** ~15% of Spaniards use digital wallets in physical stores in 2024-2025. Mobile-app payment usage grew from 4.6% (2022) → 8.2% (2023), near-doubling year-over-year.
- **Spain mobile-payment market:** projected CAGR of ~21% through 2031 (Mordor Intelligence) — value roughly doubling every 3-4 years.
- **T-Mobilitat (Barcelona):** launched Dec 2021. By mid-2024 reached ~100k weekday validations — ~7% of Barcelona metro trips. Slow rollout but on a multi-year migration plan.
- **Cash usage in Spain:** dropped from ~60% (2020-2023) to ~23% (2025) as primary payment method — roughly -10 pp / year.
- **Contactless cards:** already ~83% of card payments in 2020 (Banco de España) — that transition is mostly done.

Crucially, the lost-items corpus tracks what people **carry**, not what they **use**. Carry lags behind usage by several years because people keep cards as a backup. So the corpus will decline more slowly than payment-behaviour statistics suggest.

Per-category annual decline estimates

Category	2025 share	Driver of decline	Annual decline
Transit cards	31.6%	T-Mobilitat migration (TMB-driven)	10-15%
Bank cards	7.3%	Apple Pay / Google Pay adoption	5-8%
Wallets	16.7%	Lagging indicator of cards going digital	4-7%
Cash	0.2%	Already mostly digital	7-10% (small absolute)
IDs	15.5%	MyDNI digital ID — very early, slow rollout	2-4%
Phones	4.0%	Not displaced by digital wallets	0-1%

Category	2025 share	Driver of decline	Annual decline
Bags / clothes / misc	~24%	Not displaced	0-1%
Jewellery, glasses, keys, ...	~1%	Not displaced	0%

Projected corpus trajectory

Weighting the current 28,648-item corpus by the per-category decline rates:

Year	Estimated corpus	Cumulative decline
2025 (baseline)	28,648	—
2026	~27,000	-6%
2027	~25,400	-11%
2028	~23,500	-18%
2029	~21,500	-25%
2030	~19,200	-33%
2032	~15,000	-48%
2035	~10,500	-63%

Roughly **-6% in the first year, accelerating to -10-12% per year by 2030** as the cards/wallets backlog drains from people's pockets.

The locker network's leverage grows over time

Two effects compound:

1. **Total corpus shrinks** (~33% over five years).
2. **Locker-eligible share rises**, because the items that disappear are mostly counter-only (cards, wallets, IDs). The locker-eligible categories — bags, electronics, clothing, books — are largely unaffected.

Putting both effects together:

Year	Total items	Locker-eligible share	Locker-eligible items / year
2025	28,648	24%	~6,800
2028	~23,500	~32%	~7,500
2030	~19,200	~42%	~8,100
2035	~10,500	~60%	~6,300

The locker network has a stable target volume of **~6,000-8,000 items / year for the next decade** — but it covers an increasing fraction of TMB's total L&F operational load. Meanwhile counter operations should see *decreasing* volume.

Strategic implication

The case for investing in the locker infrastructure now strengthens as digitalisation proceeds, not weakens:

- In 2025 the locker network can lift overall returns from 18% → 27% (1.5×) — modest, because most of the corpus is counter-only.
- By 2030, the same locker network would lift overall returns from a hypothetical ~25% baseline (smaller corpus, less low-value transit-card noise) to **~45-50%** — a larger absolute and relative gain.
- By 2035, the locker network is the *primary* L&F mechanism for the remaining ~10,500-item corpus, with 60%+ of items routable to lockers.

The window to deploy and tune the network is now. By the time digitalisation has substantially shifted the corpus, having an operational, GNN-tuned locker network is what allows TMB to convert that shift into a service-quality improvement instead of just a smaller operational footprint.

Caveats on the projection

- Mobile-payment CAGR (21%) is a market-value figure, not a usage-frequency figure. Carry behaviour lags by several years.
- T-Mobilitat rollout has been slower than expected (3 years in, still single-digit share). Could accelerate or stall.
- Tourist traffic — large in Barcelona — slows the decline (visitors carry foreign physical cards).
- Senior population (T-Rosa holders) is slowest to adopt digital alternatives.
- These projections assume no major shocks (regulatory mandates, fraud scandals, card-network changes).

Sources

- Apple Pay use in Spain — Statista
- Spain Mobile Payment Market — Mordor Intelligence
- Payment methods in Spain 2025 — PaynoPain
- T-Mobilitat — TMB
- T-Mobilitat unified fare collection — ALG Global
- Cash displacement in Spain — Electronic Payments International
- Cash Use Habits 2023 — Banco de España
- Digital payments landscape in Spain — Statista

Appendix — methodology notes

- **Reclamado vs Entregat** : we use **Entregat a passatger** (18.8%) as the operational return-rate baseline. The 1.0 pp gap (5,107 vs 5,399) reflects items delivered without a formal claim entry (walk-in returns, data-entry inconsistencies).
- **"Título de transporte" interpretation**: under our assumption, this field contains registered transit titles (T-Usual, T-Jove, T-Familiar) and excludes anonymous tickets (T-Casual, T-Dia). The data does not explicitly distinguish these; staff likely don't log anonymous tickets that cannot be returned.
- **Locker eligibility list**: a working definition, not a TMB policy. The boundaries (e.g., whether wallets containing cards should be counter or locker) are operational decisions that TMB would make.

- **3x factor:** chosen as a round, defensible benchmark rather than a precise prediction. The realistic-scenario per-category breakdown in §4 lands close to this aggregate when applied only to locker-eligible items.